Yashav Bhatnagar

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Education

International Institute of Information Technology (IIIT) **Hyderabad, India**

B.Tech in Computer Science and Engineering; CGPA: 8.01 *2024 – 2028*

Relevant Coursework: Machine, Data and Learning; Advanced NLP (ongoing); Probability & Statistics; Linear Algebra; Statistical Methods in AI (self-study)

Experience

Precog Research Group, IIIT Hyderabad **Apr 2026 – Present**

Undergraduate Researcher advised by Prof. Ponnurangam Kumaraguru *Hyderabad, India*

- Investigating temporal and physical reasoning in frozen V-JEPA2 representations through layer-wise probing of intuitive physics, temporal order, recurrence, and periodic structure.
- Probing whether recurrence, period, phase, repetition count, and tempo are recoverable from intermediate representations, using temporal-shuffling and cross-context controls.

NJSecure.org (Client: Satya Vithala) **Jan 2026 – May 2026**

Software Engineer *Hyderabad, India*

- Led a 5 person team building the ORBIT Node, a cybersecurity orchestration platform for community banks, providing defense using global threat information given by a central server, with an analyst approving AI reasoned assessment to signal Wazuh (SIEM) to execute the mitigation.
- Developed the Gemini MCP Orchestrator and AI reasoning layer, atop a self-designed Neo4j asset-threat knowledge graph and co-engineered the Presidio PII risk-scoring pipeline and Wazuh telemetry integration to drive structured, continuously refreshed graph context for LLM reasoning.

Smart City Living Lab, IIIT Hyderabad **Apr 2025 – Jan 2026**

Undergraduate Researcher *Hyderabad, India*

- **Hardware Test Jig:** Engineered a Raspberry Pi-based hardware validation platform supporting seven industrial sensor protocols, exposing CLI, GUI, and web interfaces (FastAPI, Tkinter) with real-time SSE diagnostics and non-blocking hardware checks.
- Enabled rapid protocol-level debugging across heterogeneous sensor hardware, catching integration issues before deployment; work showcased at Research Fest @IIITH Spring 2026. [GitHub]

Projects

Predicting Missing Family Ties with Graph Neural Networks | *PyTorch, PyG, TransE/ComplEx/RGCN* | [GitHub]

- Implemented KG embedding models (TransE, DistMult, ComplEx, RotatE) and RGCN with leakage-aware data splitting on a 1,316-node family KG; ComplEx achieved MRR >0.87 and near-perfect Hits@10, with Louvain/Leiden community detection yielding modularity Q≈0.98 across 50 family clusters.

Audio-Guided Microbot | *ESP32-S3, Embedded C/C++, TinyML, CNNs, BLE/LoRa* | [GitHub]

- Built a fully offline, voice-controlled microbot deploying an 8-bit quantized CNN Keyword Spotting pipeline on ESP32-S3, achieving >93.87% accuracy and ∼102 ms latency, optimized with a multi-metric speech gate and BLE/LoRa.

Operating Systems & Networks Projects | *C, POSIX Threads, xv6-riscv, Socket Programming* | [xv6] [NFS] [C-Shell]

- Extended MIT's xv6-riscv kernel with three CPU schedulers (Round Robin, FCFS, CFS with vruntime-based fair scheduling) and demand-paged virtual memory backed by FIFO disk swap (lazy ELF loading, per-page dirty-bit tracking); added thread-safe `getreadcount` and `memstat` syscalls.
- Built a distributed Network File System with sentence-level locking, Git-inspired checkpointing, and asynchronous replication across horizontally scaling storage servers; and a Unix shell from scratch with pipes, I/O redirection, signal handling, and full job control.

Technical Skills

Languages: C/C++, Python, JavaScript, Assembly, SQL
AI, ML & Data Science: PyTorch, scikit-learn, NumPy, pandas, TinyML, CNNs, GNNs
Web & Databases: Node.js, Express.js, React.js, FastAPI, MongoDB, MySQL, Neo4j, Tailwind CSS, Socket.io
Systems & Embedded: POSIX Threads, Socket Programming, ESP32, Raspberry Pi, I2C/SPI/UART, BLE/LoRa
Tools & Cloud: Linux, Git, AWS, Docker, Postman, Wazuh, Presidio

Achievements

JEE Advanced 2024: Secured All India Rank **3949** among 1.8 lakh qualified candidates.
JEE Main 2024: Secured All India Rank **1189** (99.93 percentile) among 14 lakh candidates.